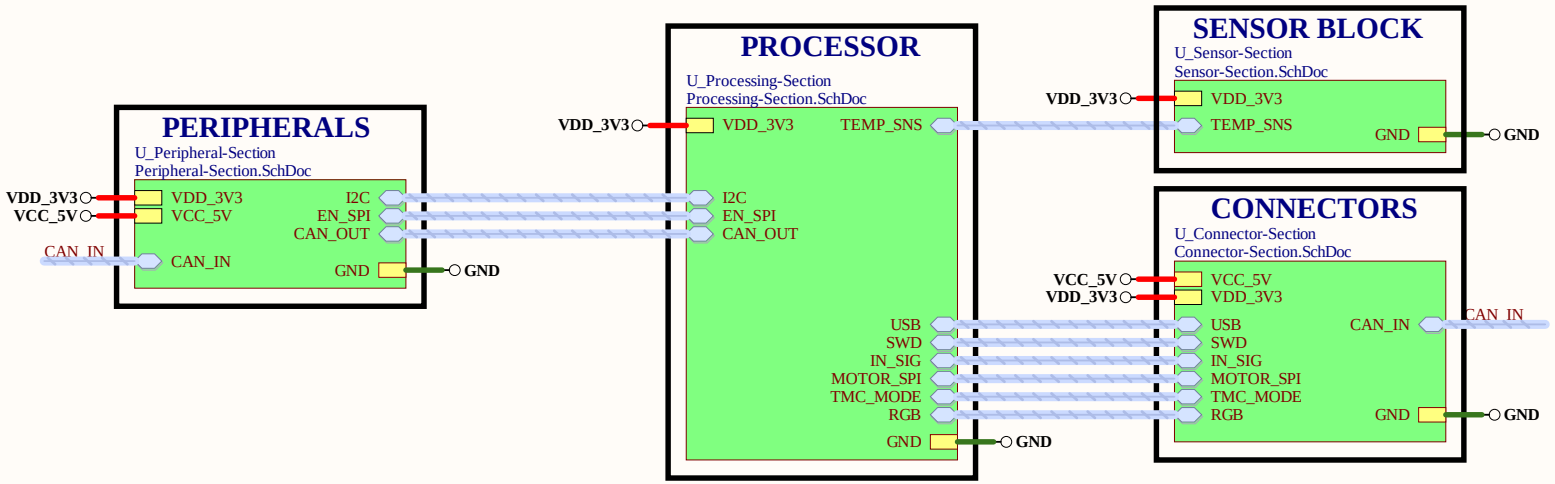
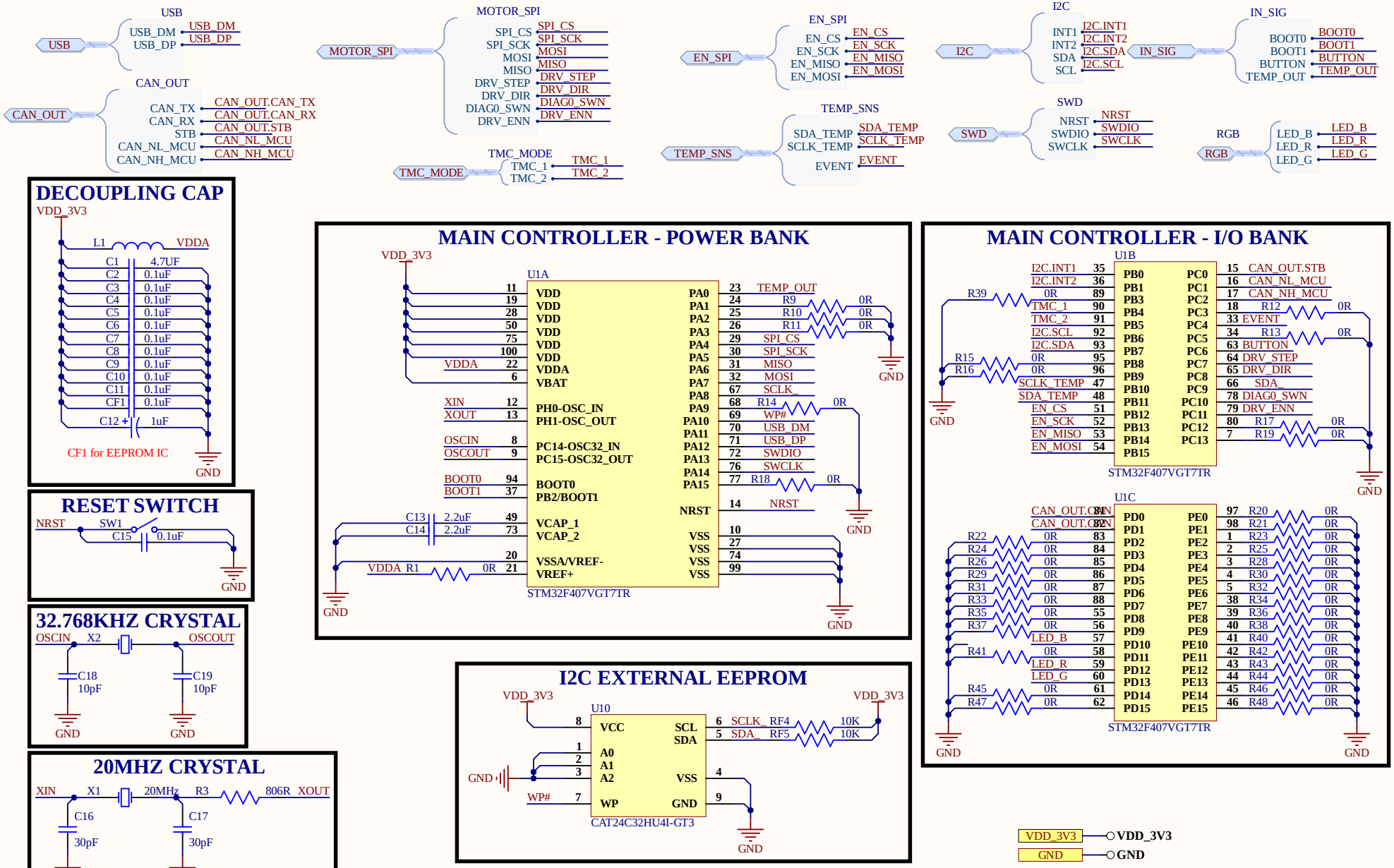
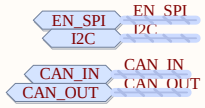
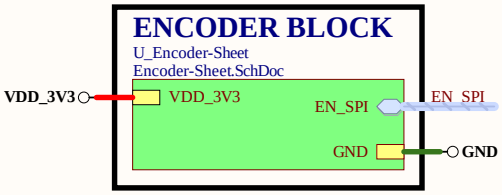
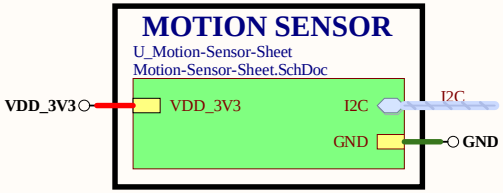
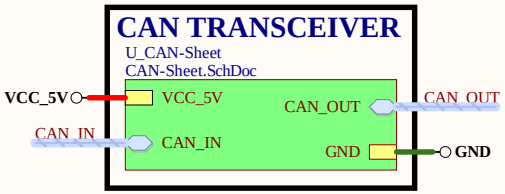

PERIPHERALS
U_Peripheral-Section
Peripheral-Section.SchDoc
VDD_3V3
VCC_5V
CAN_IN
VDD_3V3
VCC_5V
CAN_IN
I2C
EN_SPI
CAN_OUT
GND
GND
PROCESSOR
U_Processing-Section
Processing-Section.SchDoc
VDD_3V3
VDD_3V3
TEMP_SNS
I2C
EN_SPI
CAN_OUT
USB
SWD
IN_SIG
MOTOR_SPI
TMC_MODE
RGB
GND
GND
SENSOR BLOCK
U_Sensor-Section
Sensor-Section.SchDoc
VDD_3V3
VDD_3V3
TEMP_SNS
GND
GND
CONNECTORS
U_Connector-Section
Connector-Section.SchDoc
VCC_5V
VDD_3V3
VCC_5V
VDD_3V3
USB
SWD
IN_SIG
MOTOR_SPI
TMC_MODE
RGB
CAN_IN
GND
CAN_IN
GND

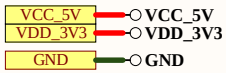

USB
USB_DM
USB_DP
CAN_OUT
CAN_TX
CAN_RX
STB
CAN_NL_MCU
CAN_NH_MCU
MOTOR_SPI
SPI_CS
SPI_SCK
MOSI
MISO
DRV_STEP
DRV_DIR
DIAG0_SWN
DRV_ENN
TMC_MODE
TMC_1
TMC_2
EN_SPI
EN_CS
EN_SCK
EN_MISO
EN_MOSI
TEMP_SNS
SDA_TEMP
SCLK_TEMP
EVENT
I2C
INT1
INT2
SDA
SCL
I2C.INT1
I2C.INT2
I2C.SDA
I2C.SCL
IN_SIG
BOOT0
BOOT1
BUTTON
TEMP_OUT
SWD
NRST
SWDIO
SWCLK
RGB
LED_B
LED_R
LED_G
DECOUPLING CAP
VDD_3V3
L1
VDDA
C1 4.7UF
C2 0.1uF
C3 0.1uF
C4 0.1uF
C5 0.1uF
C6 0.1uF
C7 0.1uF
C8 0.1uF
C9 0.1uF
C10 0.1uF
C11 0.1uF
CF1 0.1uF
C12 + 1uF
CF1 for EEPROM IC
GND
RESET SWITCH
NRST
SW1
C15 0.1uF
32.768KHZ CRYSTAL
OSCIN
X2
OSCOUT
C18 10pF
C19 10pF
20MHZ CRYSTAL
XIN
X1
20MHz
R3 806R
XOUT
C16 30pF
C17 30pF
MAIN CONTROLLER - POWER BANK
U1A
VDD
VDDA
VBAT
PH0-OSC_IN
PH1-OSC_OUT
PC14-OSC32_IN
PC15-OSC32_OUT
BOOT0
PB2/BOOT1
VCAP_1
VCAP_2
VSSA/VREF-
VREF+
PA0 TEMP_OUT
PA1 R9 0R
PA2 R10 0R
PA3 R11 0R
PA4 SPI_CS
PA5 SPI_SCK
PA6 MISO
PA7 MOSI
PA8 SCLK_
PA9 R14 0R
PA10 WP#
PA11 USB_DM
PA12 USB_DP
PA13 SWDIO
PA14 SWCLK
PA15 R18 0R
NRST
VSS
STM32F407VGT7TR
MAIN CONTROLLER - I/O BANK
U1B
U1C
PB0 PC0 CAN_OUT.STB
PB1 PC1 CAN_NL_MCU
PB3 PC2 CAN_NH_MCU
PB4 PC3 R12 0R
PB5 PC4 EVENT
PB6 PC5 R13 0R
PB7 PC6 BUTTON
PB8 PC7 DRV_STEP
PB9 PC8 DRV_DIR
PB10 PC9 SDA_
PB11 PC10 DIAG0_SWN
PB12 PC11 DRV_ENN
PB13 PC12 R17 0R
PB14 PC13 R19 0R
PB15
PD0 PE0 R20 0R
PD1 PE1 R21 0R
PD2 PE2 R23 0R
PD3 PE3 R25 0R
PD4 PE4 R28 0R
PD5 PE5 R30 0R
PD6 PE6 R32 0R
PD7 PE7 R34 0R
PD8 PE8 R36 0R
PD9 PE9 R38 0R
PD10 PE10 R40 0R
PD11 PE11 R42 0R
PD12 PE12 R43 0R
PD13 PE13 R44 0R
PD14 PE14 R46 0R
PD15 PE15 R48 0R
I2C EXTERNAL EEPROM
U10
VCC
A0
A1
A2
WP
SCL
SDA
VSS
GND
SCLK_ RF4 10K
SDA_ RF5 10K
WP#
CAT24C32HU4I-GT3
VDD_3V3
GND

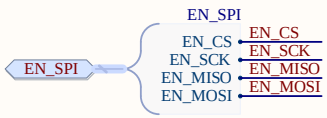

EN_SPI
I2C
EN_SPI
I2C
CAN_IN
CAN_OUT
CAN_IN
CAN_OUT


ENCODER BLOCK
U_Encoder-Sheet
Encoder-Sheet.SchDoc
VDD_3V3
VDD_3V3
EN_SPI
EN_SPI
GND
GND

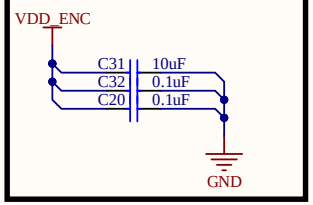

MOTION SENSOR
U_Motion-Sensor-Sheet
Motion-Sensor-Sheet.SchDoc
VDD_3V3
VDD_3V3
I2C
I2C
GND
GND


CAN TRANSCEIVER
U_CAN-Sheet
CAN-Sheet.SchDoc
VCC_5V
VCC_5V
CAN_OUT
CAN_OUT
CAN_IN
CAN_IN
GND
GND

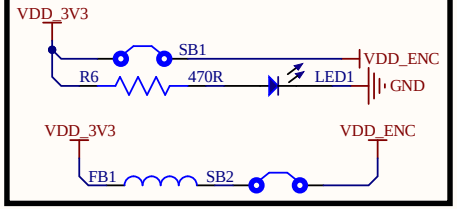

VCC_5V
VDD_3V3
GND
VCC_5V
VDD_3V3
GND



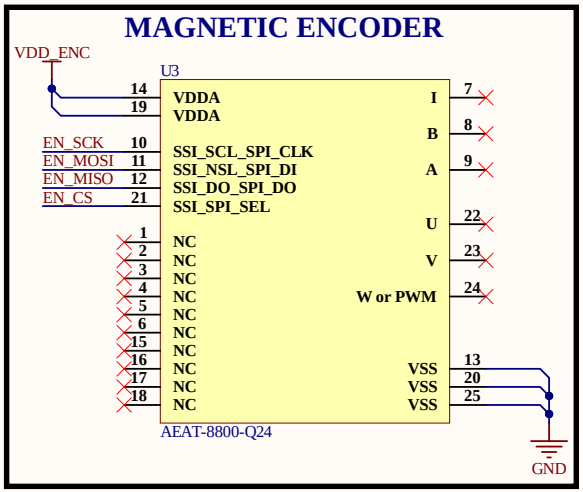
DECOUPLING CAPS



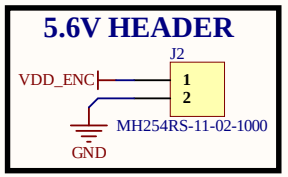
SUPPORTING CIRCUIT



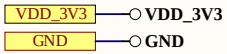
MAGNETIC ENCODER

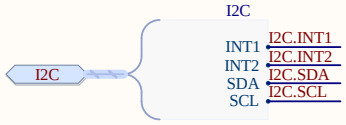


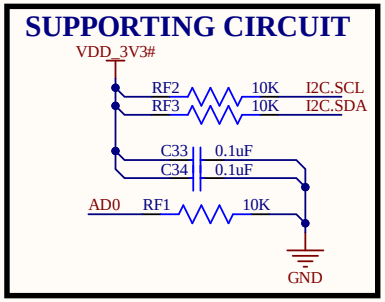
5.6V HEADER

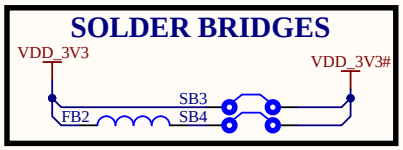


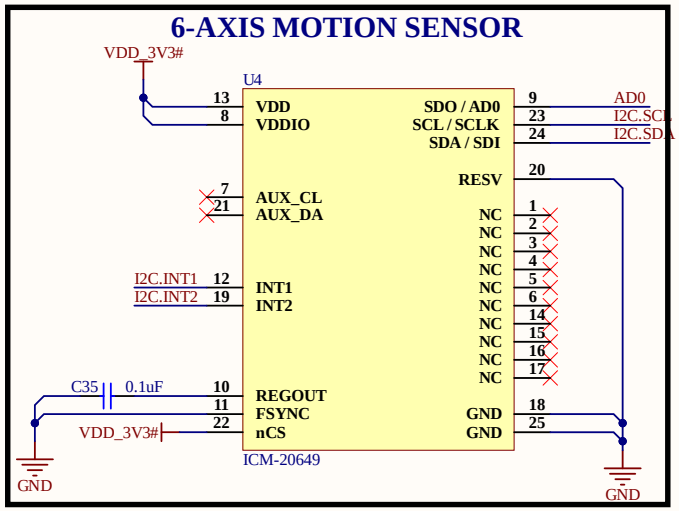
This header will be used to provide external power (5.6V) to Encoder in order to program its OTP Registers. For this, SB1 bridge needs to be disconnected.

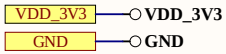


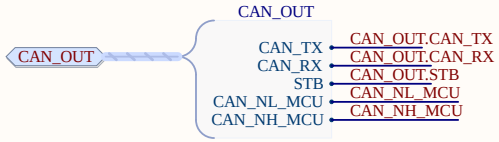

I2C
I2C
INT1
INT2
SDA
SCL
I2C.INT1
I2C.INT2
I2C.SDA
I2C.SCL

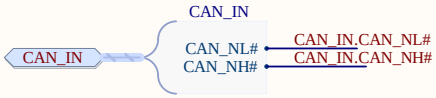

SUPPORTING CIRCUIT
VDD_3V3#
RF2
10K
I2C.SCL
RF3
10K
I2C.SDA
C33
0.1uF
C34
0.1uF
AD0
RF1
10K
GND


SOLDER BRIDGES
VDD_3V3
VDD_3V3#
SB3
FB2
SB4

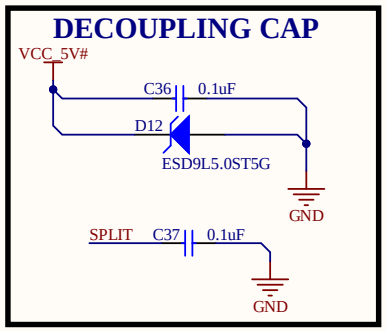

6-AXIS MOTION SENSOR
VDD_3V3#
U4
13
VDD
8
VDDIO
7
AUX_CL
21
AUX_DA
I2C.INT1
12
INT1
I2C.INT2
19
INT2
C35
0.1uF
10
REGOUT
11
FSYNC
22
nCS
VDD_3V3#
GND
ICM-20649
SDO / AD0
SCL / SCLK
SDA / SDI
RESV
NC
9
AD0
23
I2C.SCL
24
I2C.SDA
20
1
2
3
4
5
6
14
15
16
17
18
25
GND
GND

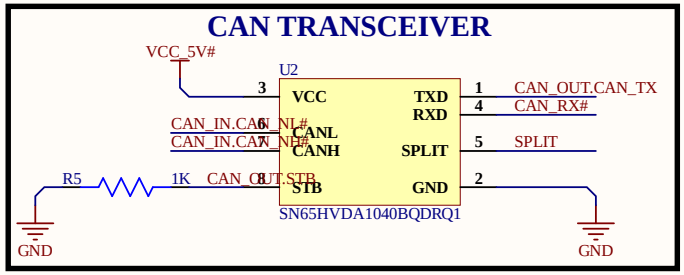

VDD_3V3
VDD_3V3
GND
GND

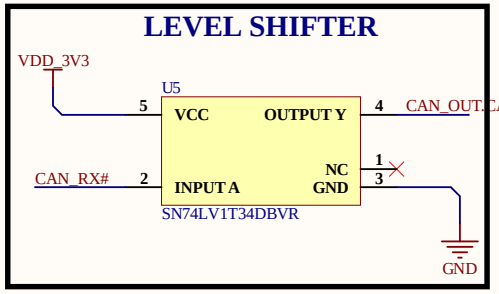


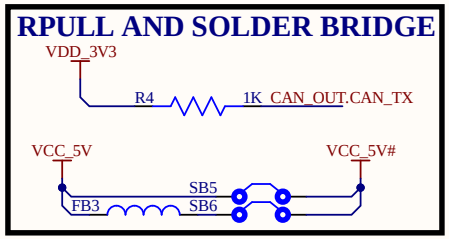


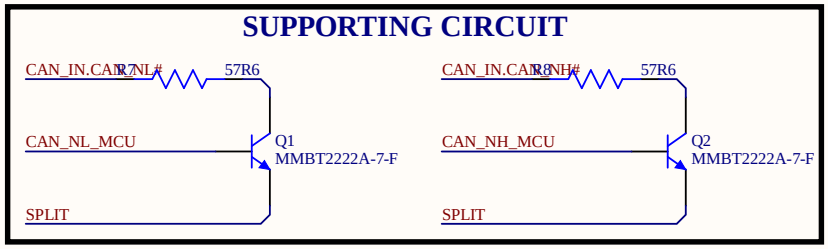
Connections for CAN are w.r.t Controller Rx and Tx

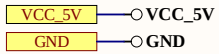


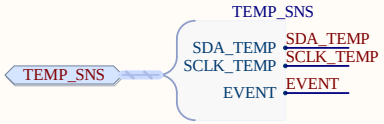


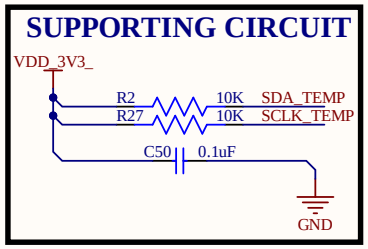


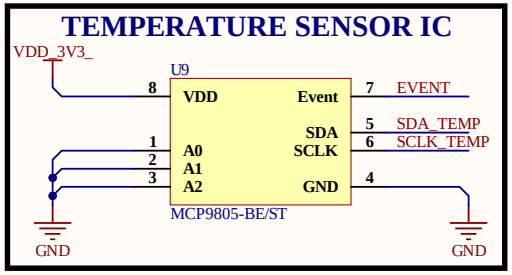


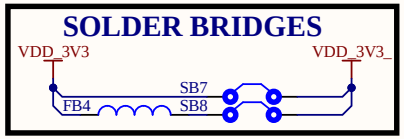


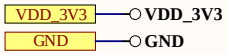



TEMP_SNS
TEMP_SNS
SDA_TEMP
SCLK_TEMP
EVENT
SDA_TEMP
SCLK_TEMP
EVENT

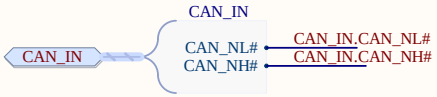

SUPPORTING CIRCUIT
VDD_3V3_
R2
10K
SDA_TEMP
R27
10K
SCLK_TEMP
C50
0.1uF
GND

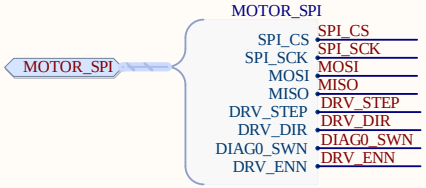

TEMPERATURE SENSOR IC
VDD_3V3_
U9
8
VDD
Event
7
EVENT
SDA
5
SDA_TEMP
1
A0
SCLK
6
SCLK_TEMP
2
A1
3
A2
GND
4
MCP9805-BE/ST
GND
GND


SOLDER BRIDGES
VDD_3V3
VDD_3V3_
SB7
FB4
SB8

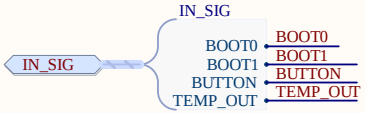

VDD_3V3
VDD_3V3
GND
GND


USB
USB_DM
USB_DP
USB_DM
USB_DP
USB

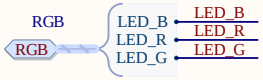

CAN_IN
CAN_NL#
CAN_NH#
CAN_IN.CAN_NL#
CAN_IN.CAN_NH#
CAN_IN

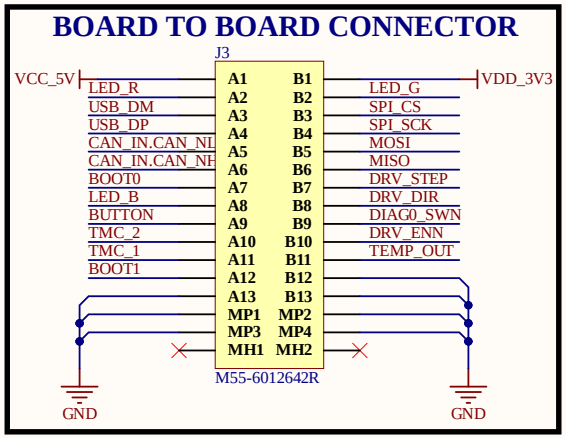

MOTOR_SPI
SPI_CS
SPI_SCK
MOSI
MISO
DRV_STEP
DRV_DIR
DIAG0_SWN
DRV_ENN
SPI_CS
SPI_SCK
MOSI
MISO
DRV_STEP
DRV_DIR
DIAG0_SWN
DRV_ENN
MOTOR_SPI

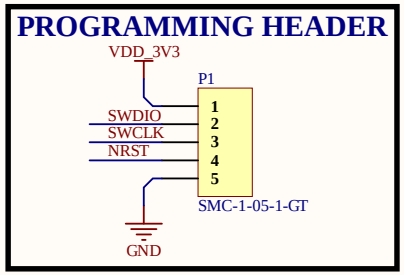

TMC_MODE
TMC_1
TMC_2
TMC_1
TMC_2
TMC_MODE

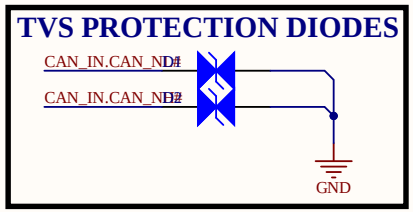

IN_SIG
BOOT0
BOOT1
BUTTON
TEMP_OUT
BOOT0
BOOT1
BUTTON
TEMP_OUT
IN_SIG

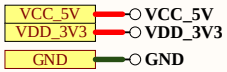

SWD
NRST
SWDIO
SWCLK
NRST
SWDIO
SWCLK
SWD


RGB
LED_B
LED_R
LED_G
LED_B
LED_R
LED_G
RGB


BOARD TO BOARD CONNECTOR
J3
VCC_5V
LED_R
USB_DM
USB_DP
CAN_IN.CAN_NL
CAN_IN.CAN_NH
BOOT0
LED_B
BUTTON
TMC_2
TMC_1
BOOT1
A1 A2 A3 A4 A5 A6 A7 A8 A9 A10 A11 A12 A13 MP1 MP3 MH1
B1 B2 B3 B4 B5 B6 B7 B8 B9 B10 B11 B12 B13 MP2 MP4 MH2
LED_G
SPI_CS
SPI_SCK
MOSI
MISO
DRV_STEP
DRV_DIR
DIAG0_SWN
DRV_ENN
TEMP_OUT
VDD_3V3
M55-6012642R
GND
GND


PROGRAMMING HEADER
VDD_3V3
P1
SWDIO
SWCLK
NRST
1
2
3
4
5
SMC-1-05-1-GT
GND


TVS PROTECTION DIODES
CAN_IN.CAN_NL#
CAN_IN.CAN_NH#
GND


VCC_5V
VDD_3V3
GND
VCC_5V
VDD_3V3
GND